

# Autodesk® **Scaleform**®

## Scaleform

Scaleform 4.0

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: 1.01  
: 2012 6 19

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## 차례

소개

사운드 시스템 초기화

비디오 사운드 시스템 초기화

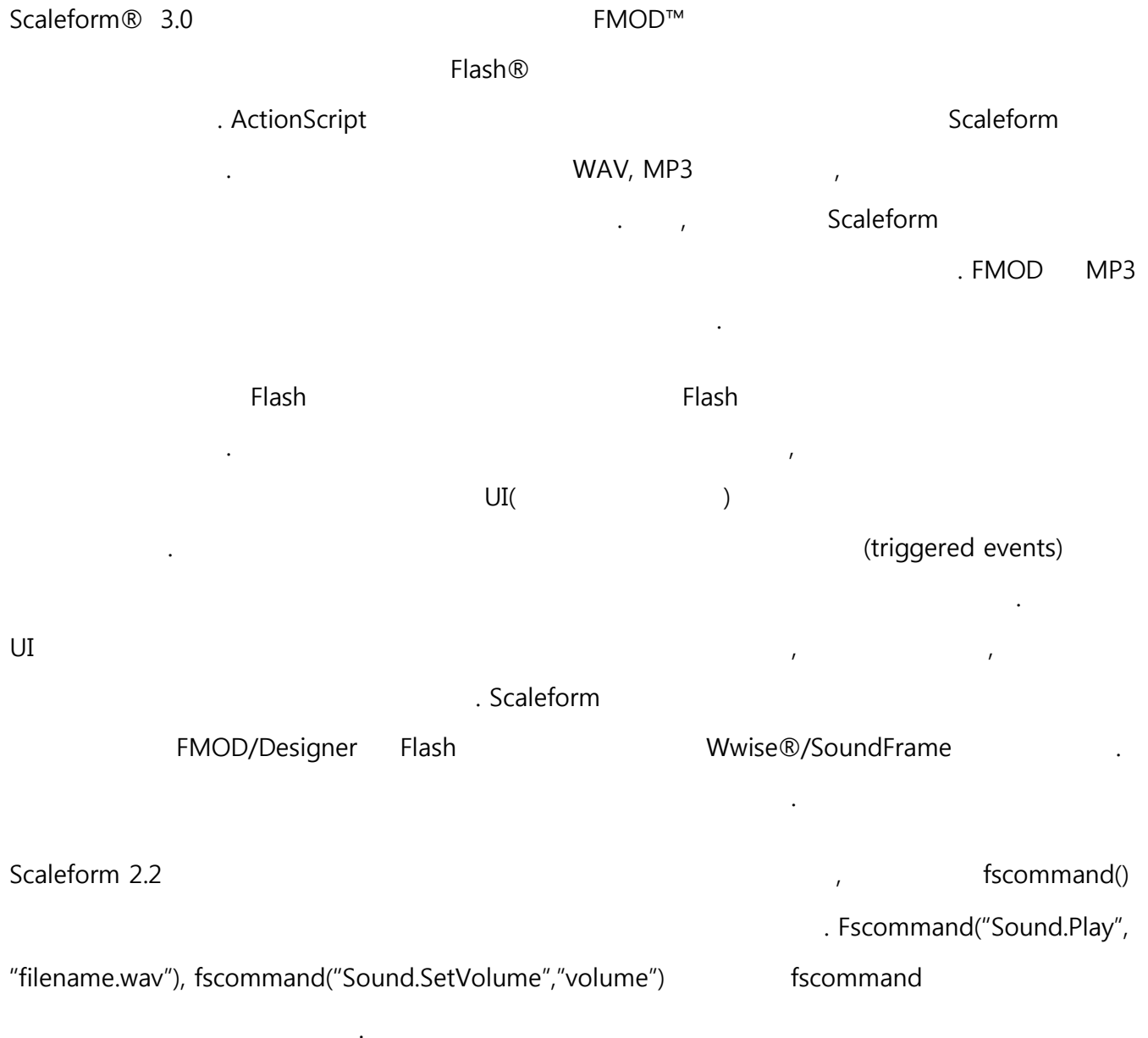
사운드 이벤트 인터페이스

비디오 사운드 시스템 인터페이스

샘플 소스 코드

추가 자료

# 소개



## 사운드 시스템 초기화

```

graph TD
    Scaleform --> GfxAudio[Gfx::Audio]
    GfxAudio --> GfxLoader[Gfx::Loader]
    GfxLoader --> SWF
    SWF --> SoundRenderer[Sound::SoundRenderer]
    SoundRenderer --> Cplusplus[C++]
    Cplusplus --> Scaleform
    Scaleform --> FMOD
    FMOD --> SoundRenderer
    SoundRenderer --> Scaleform

```

```

:
SoundRenderer* pSoundRenderer = GetSoundRenderer();
if (pSoundRenderer)
{
    // Sound renderer and parameters for playing SWF streaming sounds
    // are provided to the loader by Gfx::Audio. These parameters
    // control when and how the synchronization between SWF frames
    // and sound frames should be done.
    Ptr<Gfx::Audio> pAudioState = *new Gfx::Audio(pSoundRenderer);
    loader.SetAudio(pAudioState);
    pSoundRenderer->SetMasterVolume(SoundVolume);
    pSoundRenderer->Mute(IsMute);
}

```

Flash .

Wwise .

Scaleform Player Flash .

## 비디오 사운드 시스템 초기화

```

                                                                    Gfx::Video::VideoSoundSystem
                                                                    . Scaleform
Gfx::Video::Video
Sound::SoundRenderer
:
SoundRenderer* pSoundRenderer = GetSoundRenderer();
if (pSoundRenderer)
{
    // Setting a video sound system instance which is based on
    // SoundRenderer interface
    pVideo->SetSoundSystem(pSoundRenderer);
}

:
// Setting a video sound system instance which is specific to
// a particular platform
pVideo->SetSoundSystem(
    Ptr<Video::VideoSoundSystem>(*new Video::VideoSoundSystemDX8(0)));

SetSoundSystem()          Gfx::Video::Video
                                                                    .
```

## 사운드 이벤트 인터페이스

Scaleform 3.2 FMOD/Designer Flash  
Wwise®/SoundFrame(Sound::SoundEvent::PostEvent Sound::SoundEvent::SetParam)  
.  
ActionScript  
Scaleform Flash player  
.  
Scaleform  
.

1. Sound::SoundEventFMOD - FMOD Ex API
2. Sound::SoundEventFMODDesigner - FMOD EventSystem API , FMOD Designer
3. Sound::SoundEventWwise - Wwise API
4. Sound::SoundEventWwiseSF - Wwise SoundFrame

```

#include "Sound/Sound_SoundEventWwise.h"
Ptr<Sound::SoundEvent> soundEvent = *new Sound::SoundEventWwiseSF;

soundEvent->PostEvent("Sound_PostEvent", "Play_Hello");
soundEvent->SetParam("Sound_SetParam", "Enable_Effect", 85);

```

Scaleform Player , Flash



# 비디오 사운드 시스템 인터페이스

Gfx::Video::VideoSoundSystem    Gfx::Video::Video

Video::SetSoundSystem()

Gfx::Video

- Video::VideoSoundSystemDX8    –    Windows    DirectSound
- Video::VideoSoundSystemXA2    –    Windows    Xbox360    XAudio2
- Video::VideoSoundSystemPS3    –    PS3    MultiStream
- Video::VideoSoundSystemWii    –    Wii
- Video::VideoSoundSystemFMOD - FMOD
- Video::VideoSoundSystemWwise – Wwise

Scaleform

Scaleform

Sound::SoundRenderer

Gfx::Video::VideoSoundSystem

Video::VideoSoundSystem

Video::VideoSound

Sound::SoundRenderer

Sound::SoundRenderer

Video::Video

.(

.)

:

`#include "Video/Video_VideoSoundSystemXA2.h"`

`pvc->SetSoundSystem(Ptr<Video::VideoSoundSystem>(*new VideoSoundSystemXA2(0, 0)));`

```

        Video::VideoSoundSystem  Video::VideoSound
        VideoSoundSystem
VideoSoundSystem  VideoSound
        , Create  . Scaleform
        . (  .)  VideoSound
        , Scaleform
        VideoSound::PCMStream
        VideoSoundSystem
        .
        VideoSoundSystemWwise  Wwise SDK  v2009.2.1 build 3271
        . Audiokinetic
        /
        SDK\samples\Plugins\AkAudioInput  Wwise
        . Scaleform Video  Wwise SDK
        .
        :
        #include "Video/Video_VideoSoundSystemWwise.h"
        Ptr<Video::VideoSoundSystem> wwiseSound = Ptr<Video::VideoSoundSystem>(*new
            VideoSoundSystemWwise());
        pVideo->SetSoundSystem(wwiseSound);
        wwiseSound->Update();

        Scaleform Player
        . ,
        Getting Started with Video
    
```

## 예제 소스 코드

- Scaleform Player , FMOD/Wwise :
  - o Apps\Samples\FxPlayer\FxPlayer.cpp
  - o Apps\Samples\Common\FxSoundFMOD.cpp
  - o Apps\Samples\Common\FxSoundWwise.cpp
- FMOD :
  - o Src\Sound\Sound\_SoundRendererFMOD.h
  - o Src\Sound\Sound\_SoundRendererFMOD.cpp
- (FMOD/Designer Wwise/SoundFrame):
  - o Src\Sound\Sound\_SoundEventFMOD.h
  - o Src\Sound\Sound\_SoundEventWwise.h
- :
  - o Src\Video\Video\_VideoSoundSystem\*.cpp
- Flash :
  - o Bin\Data\AS2\Samples\Audio\_Demo\_AS2 fla
  - o Bin\Data\AS3\Samples\AudioDemo\_AS3 fla

## 추가 자료

CRI Movie ,  
(export)

ActionScript

SWF

- [Adobe Flash ActionScript 2.0](#)
- [Scaleform - Flash Support Overview](#)
- [Getting Started with Video](#)
- [CRI Movie Encoder Command Line Tool Overview](#)
- [GFxExport Reference Guide](#)